

Norway

SLEIPNER

Volve F

F-11

15/9-F-11 T2

Plan: 15/9-F-11 T2 rev0

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 T2	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 T2 rev0	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9		
Site Position:		Northing:	6,478,563.53 m
From:	Map	Easting:	435,050.02 m
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in
		Latitude:	58° 26' 29.8068 N
		Longitude:	1° 53' 14.9285 E
		Grid Convergence:	-0.95 °

Well	F-11		
Well Position	+N/-S	4.65 m	Northing:
	+E/-W	-0.93 m	Easting:
Position Uncertainty	0.00 m		Wellhead Depth:
			91.00 m
			Latitude:
			58° 26' 29.9566 N
			Longitude:
			1° 53' 14.8667 E
			Water Depth:
			91.00 m

Wellbore	15/9-F-11 T2				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MAGNETICREFERENCE	21.03.2013	-1.19	71.67	50,486

Design	15/9-F-11 T2 rev0				
Audit Notes:					
Version:	Phase:	PLAN	Tie On Depth:	202.00	
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	4.65	-0.93	62.38	

Survey Tool Program	Date	05.01.2016			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
147.70	193.00	F-11 Gyro SDC (15/9-F-11)	Gyro, old	Other Gyro Tools	
202.00	202.00	F-11 Gyro SDC 202-306m after reaming (1	Gyro, old	Other Gyro Tools	
202.00	650.00	15/9-F-11 T2 rev0 (15/9-F-11 T2)	NOT_useMWD gyro, Baker Hughes' MWD gyro service (GyroTrak)		
650.00	1,368.00	15/9-F-11 T2 rev0 (15/9-F-11 T2)	Magnetic, IFR, mag-corr	Magnetic Tools (MWD, EMS)	
1,368.00	2,570.00	15/9-F-11 T2 rev0 (15/9-F-11 T2)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)	
2,570.00	4,550.81	15/9-F-11 T2 rev0 (15/9-F-11 T2)	Magn, IFR, non-mag, dual in	Magnetic Tools (MWD, EMS)	

Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
202.00	0.26	233.95	202.00	4.49	-1.11	-0.24	0.000	0.000	0.000	
209.00	0.26	233.95	209.00	4.47	-1.13	-0.27	0.000	0.000	0.000	
243.67	1.00	355.00	243.67	4.73	-1.22	-0.23	1.000	0.640	104.735	
423.67	1.00	355.00	423.64	7.86	-1.50	0.98	0.000	0.000	0.000	
473.02	0.90	188.09	472.99	7.90	-1.59	0.92	1.150	-0.058	-101.467	
565.55	0.90	188.09	565.51	6.45	-1.80	0.07	0.000	0.000	0.000	

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 T2	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 T2 rev0	Database:	Production EDM P246N

Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
745.90	6.67	290.00	745.45	8.63	-11.85	-7.83	1.150	0.959	16.953	
896.73	12.80	262.41	894.11	9.42	-36.68	-29.47	1.500	1.220	-5.488	
1,077.10	12.80	262.41	1,070.00	4.14	-76.30	-67.02	0.000	0.000	0.000	
1,257.28	27.80	264.55	1,238.50	-2.53	-138.27	-125.03	2.500	2.497	0.357	
1,797.15	27.80	264.55	1,716.06	-26.46	-388.93	-358.22	0.000	0.000	0.000	
2,567.53	37.00	69.60	2,442.00	44.90	-345.90	-287.01	2.500	0.358	6.427	
2,577.53	37.00	69.60	2,449.99	47.00	-340.26	-281.04	0.000	0.000	0.000	
3,009.43	80.19	69.60	2,669.75	172.47	-2.87	76.07	3.000	3.000	0.000	
3,708.79	80.19	69.60	2,788.91	412.68	643.04	759.75	0.000	0.000	0.000	
4,355.15	14.18	69.76	3,204.30	568.48	1,063.45	1,204.47	3.064	-3.064	0.008	
4,550.81	14.18	69.76	3,394.00	585.06	1,108.42	1,252.01	0.000	0.000	0.000	

Design Targets										
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude	
- hit/miss target										
- Shape										
F-15 D N Pilot Top Hugir	0.00	360.00	3,204.30	568.48	1,063.45	6,479,131.80	436,113.10	58° 26' 48.7409 N	1° 54' 19.8979 E	
- plan misses target center by 0.27m at 4355.15m MD (3204.30 TVD, 568.73 N, 1063.36 E)										
- Rectangle (sides W100.00 H50.00 D0.00)										

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
202.00	202.00	30"	30.000	36.000	
1,358.00	1,327.60	20"	20.000	26.000	
2,571.00	2,444.77	14"	14.000	17.500	

Checked By: _____ Approved By: _____ Date: _____